

Population Effect Estimation in Bayesian Hierarchical Models Using Sum-to-Zero Constraints

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Slides and code at



github.com/spinkney/open-talks

Agenda

- Preliminaries
- What is a sum-to-zero constraint?
- Hierarchical models between frequentist and bayesian approaches
- How is it possible to translate between them?
- Implications

Preliminaries: Distributions

We will restrict ourselves to elliptically symmetric distributions for hierarchical models, i.e.,

$$f(x) = c g(x^\top \Sigma_x^{-1} x),$$

$$\int_{\mathbb{R}^d} f(x) dx = 1$$

where¹

- the generator $g(\cdot) : \mathbb{R}^+ \rightarrow \mathbb{R}^+$
- the normalizing constant c ensures $f(x)$ integrates to 1

Preliminaries: Distributions

$$f(x) = c g(x^\top \Sigma_x^{-1} x),$$

$$\int_{\mathbb{R}^d} f(x) dx = 1$$

The choice of g controls the tail behavior, allowing both light- and heavy-tailed priors within the same family:

- Normal: $g(t) \propto e^{-t/2}$
- Student- t_ν (Cauchy at $\nu = 1$): $g(t) \propto (1 + t/\nu)^{-(\nu+d)/2}$
- Logistic: $g(t) \propto e^{-t} / (1 + e^{-t})^2$
- Laplace: $g(t) \propto e^{-\sqrt{t}}$

What is a sum-to-zero constraint?

It's simply

$$\sum_j^n a_j = 0$$

To construct one might randomly draw i.i.d. $n - 1$ values y and

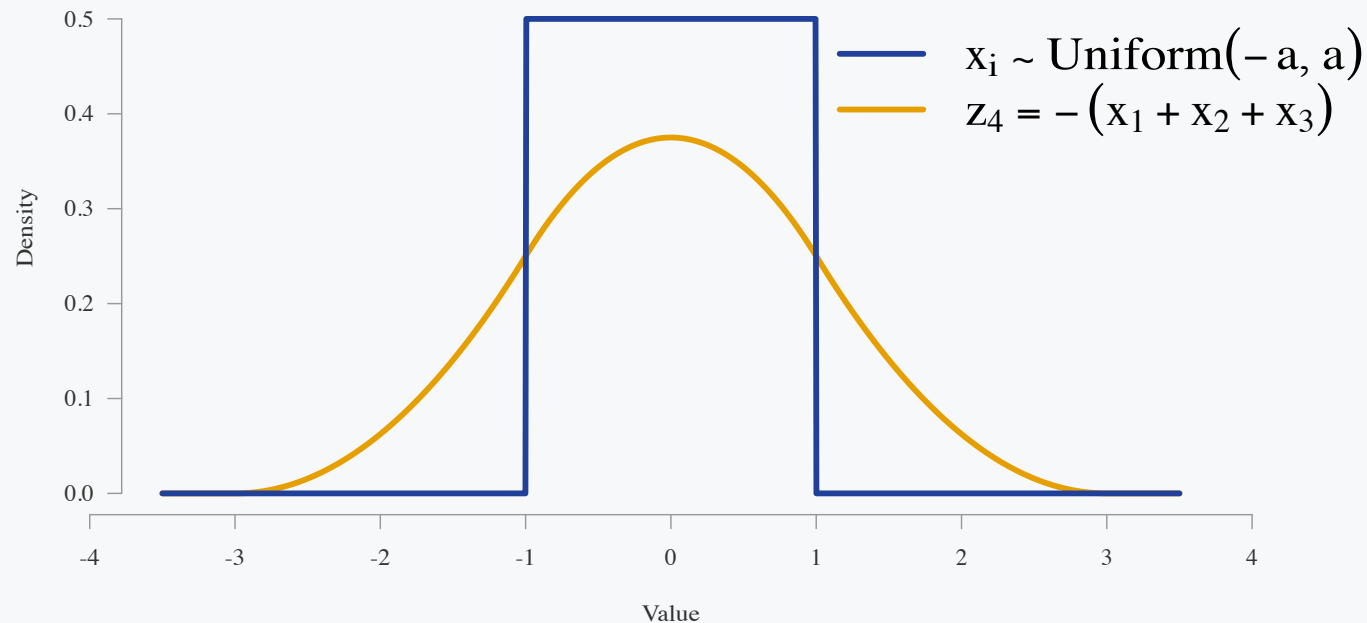
$$y_n \stackrel{\text{set}}{=} - \sum_i^{n-1} y_i$$

$$\mathbf{z} = [y_1 \ y_2 \ \dots \ y_{n-1} \ y_n]$$

$$\sum_i^n z_i = 0$$

What is a sum-to-zero constraint?

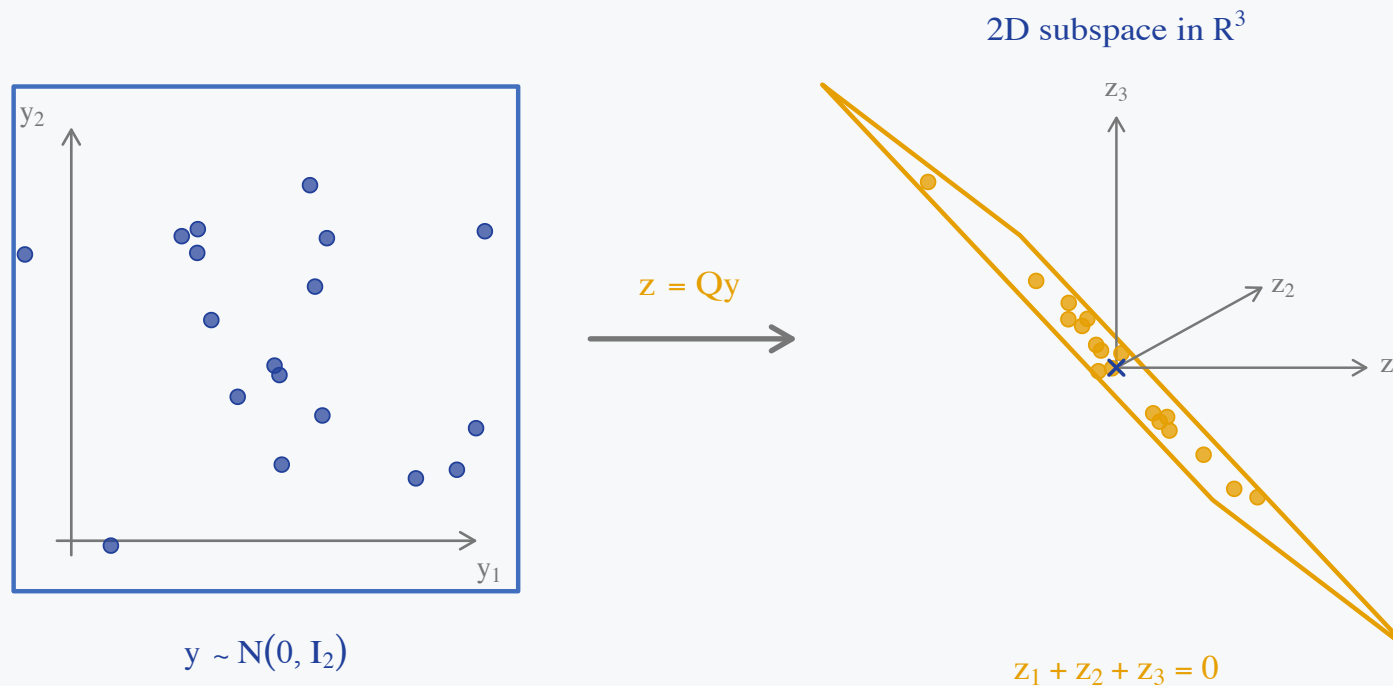
Since only z_n is a function of y the marginal density will be the density of the sum of $z_{i:n-1}$. Most modelers would not want this!



The implicit prior on the last element is clearly different. Furthermore, the covariance of the final element is coupled strongly with the other elements.

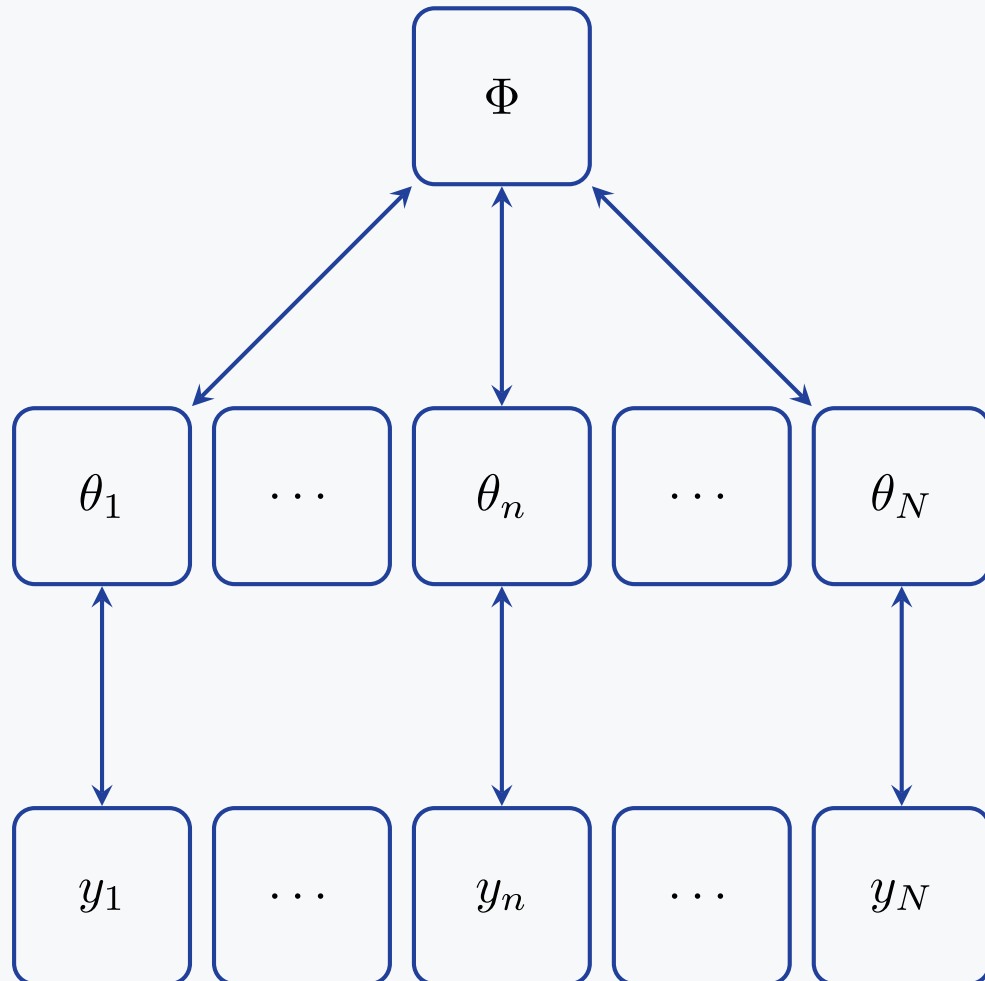
What is a sum-to-zero constraint?

We can balance the density across the values by using an orthonormal basis for the zero-sum subspace, $QQ^\top = P_0 = I_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top$.



All the values are now isotropic and exchangeable.

Preliminaries: Background on Bayesian hierarchical models



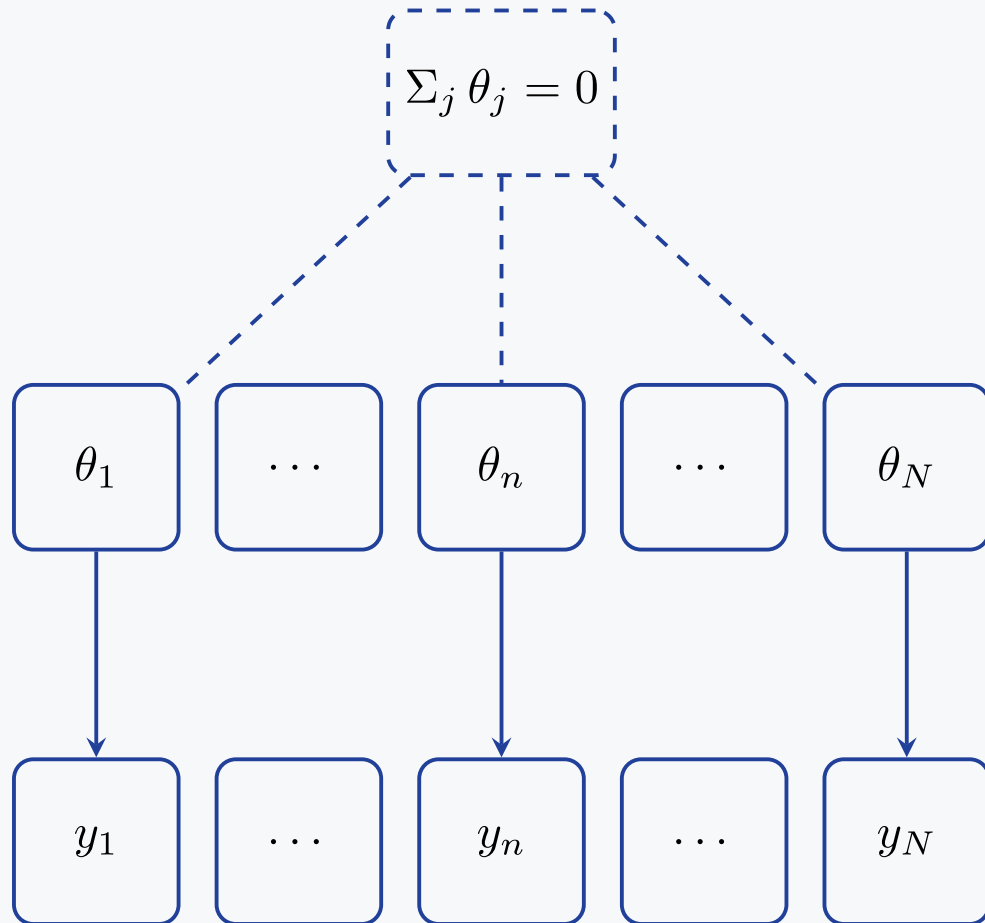
Sharing of information happens

- Φ population distribution
- Globally
- Bi-directionally
- Partial pooling

When the evidence or data for a parameter are

- low
 - estimate is closer to prior
- large
 - data swamps prior
 - prior pull is weak

Preliminaries: Background on frequentist hierarchical models



No population distribution Φ

- the θ_j are **fixed unknowns**, not draws
- no pooling: each θ_j is estimated from group j data alone
- information flows one way, $\theta_j \rightarrow y_j$

Groups are tied together only by a *hard* constraint

- e.g. effects coding $\sum_j \theta_j = 0$
- identifies the grand mean instead of a population parameter

What is the population effect?

Let's take a 2-level, varying slopes, varying intercept model with i units and j groups

$$y_{ij} = \underbrace{\alpha + a_j}_{\text{varying intercept}} + X \underbrace{(\beta + b_j)}_{\text{varying slope}} + \epsilon_{ij}$$

population parameters

The diagram illustrates the structure of a 2-level, varying slopes, varying intercept model. At the top, the text 'population parameters' is centered. Two blue arrows point downwards from this text to the terms $\alpha + a_j$ and $\beta + b_j$ in the equation below. The equation is $y_{ij} = \alpha + a_j + X(\beta + b_j) + \epsilon_{ij}$. A blue bracket is placed under $\alpha + a_j$ with the label 'varying intercept' below it. Another blue bracket is placed under $\beta + b_j$ with the label 'varying slope' below it.

Group-level (conditional): $E(y_{ij} \mid X, j) = \alpha + a_j + X(\beta + b_j)$

Population-level (marginal): $E(y_{ij} \mid X) = \alpha + X\beta$

when $E(a_j) = E(b_j) = 0$

Different populations

The data inform the group coefficients α_j and β_j whereas the population α and β sit one level up.

Bayesian hierarchical model

α and β are *population* parameters and the observed groups are **draws from** that population

$$a_j \sim \mathcal{N}(\alpha, \sigma_a), \quad b_j \sim \mathcal{N}(\beta, \sigma_b)$$

A benefit is that new groups may still be inferred from the population distribution as any finite draw of J groups has

$$\bar{a} = \frac{1}{J} \sum_j a_j \neq \alpha$$

Frequentist effects coding

No population to draw from — the J observed groups **are** the population

The “population” parameter is the finite-population grand mean

$$\alpha = \frac{1}{J} \sum_j (\alpha + a_j) \iff \sum_j a_j = 0$$

Is there an equivalence?

Suppose we restrict to varying intercepts, $x_i \sim \mathcal{N}(\alpha + a_j, \sigma_x)$:

Bayesian hierarchical model

$$a_j^{\text{bayes}} \sim \mathcal{N}(0, \sigma_a)$$

$$\bar{a} = \frac{1}{J} \sum_j a_j^{\text{bayes}} \sim \mathcal{N}\left(0, \frac{\sigma_a^2}{J}\right)$$

Frequentist effects coding

$$\alpha = \frac{1}{J} \sum_j (\alpha + a_j^{\text{s2z}}) \iff \sum_j a_j^{\text{s2z}} = 0$$

Take

$$\alpha_{\text{s2z}} \stackrel{\text{set}}{=} \alpha_{\text{bayes}} + \bar{a}$$

$$a_j^{\text{s2z}} \stackrel{\text{set}}{=} a_j^{\text{bayes}} - \bar{a}$$

$$\sum a_j^{\text{s2z}} = 0$$

Does it work?

Let's simulate six deliberately unbalanced groups from

$$x_i \sim \mathcal{N}(\alpha + a_{j[i]}, \sigma_x), \quad a_j \sim \mathcal{N}(0, \tau^2),$$

with

```
alpha = 5  
tau = 2.5  
sigma_x = 1  
n_j = (8, 13, 21, 34, 55, 89)
```

For this finite draw,

$$\frac{1}{J} \sum_j a_j^{\text{bayes}} = \bar{a} = 0.85, \quad \alpha_{s2z} = 5.85.$$

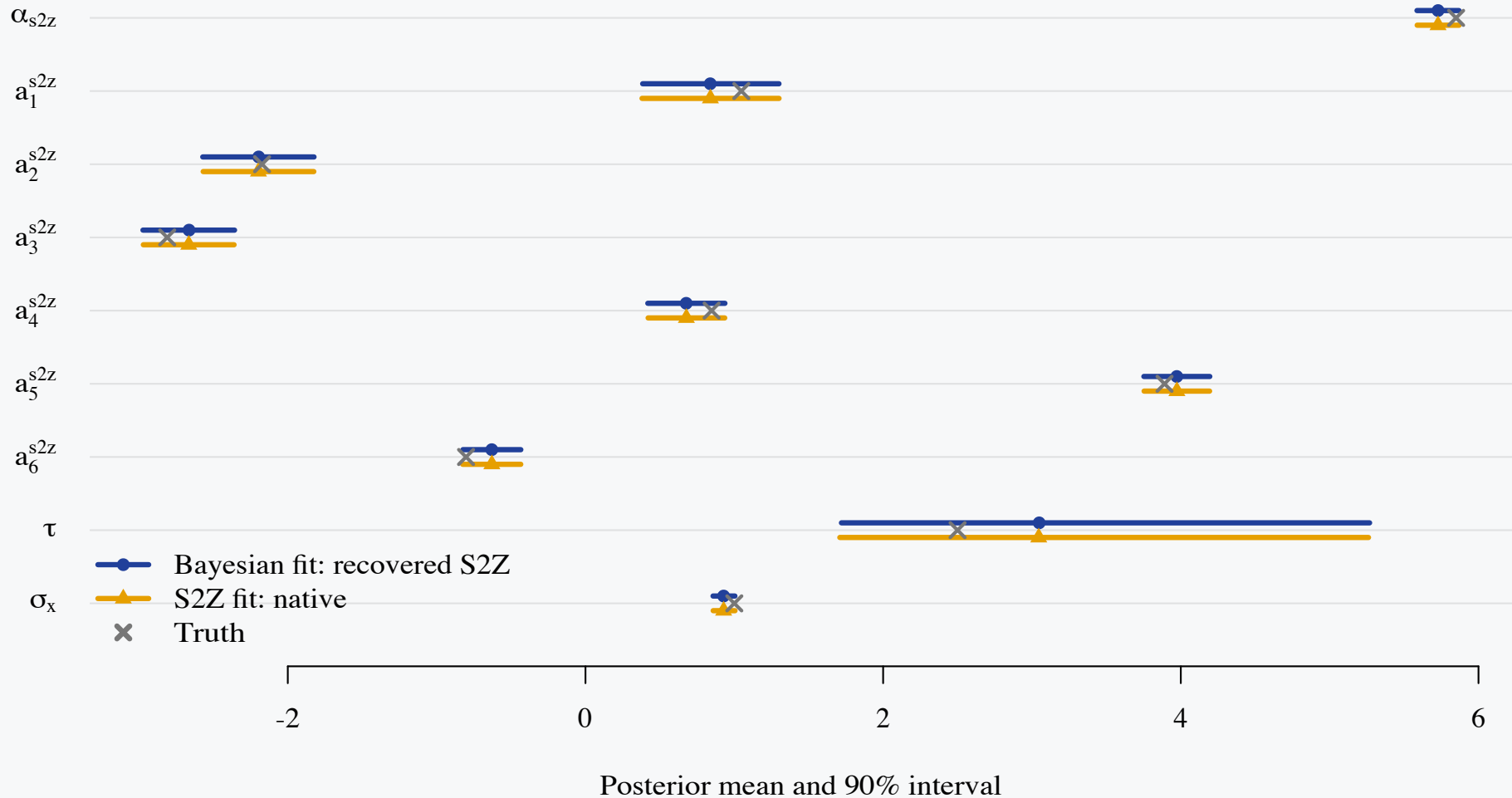
Sampling 10k iterations over 4 chains. The first Stan program samples unconstrained Bayesian varying effects; the second samples a

Conversion

Bayesian specification and conversion to sum-to-zero equivalent

```
1 data {
2   int<lower=1> N;
3   int<lower=2> J;
4   array[N] int<lower=1, upper=J> group;
5   vector[N] x;
6   real<lower=0> alpha_prior_sd;
7 }
8 parameters {
9   real alpha_bayes;
10  vector[J] a_bayes;
11  real<lower=0> tau;
12  real<lower=0> sigma_x;
13 }
14 model {
15   alpha_bayes ~ normal(0, alpha_prior_sd);
16   a_bayes ~ normal(0, tau);
17   tau ~ normal(0, 5);
18   sigma_x ~ normal(0, 2);
19   x ~ normal(alpha_bayes + a_bayes[group], sigma_x);
20 }
21 generated quantities {
22   real mean_a = mean(a_bayes);
23   real alpha_s2z_recovered = alpha_bayes + mean_a;
24   vector[J] a_s2z_recovered = a_bayes - mean_a;
25 }
```

The two fits agree



Across α_{s2z} and the six a_j^{s2z} , the largest posterior-mean difference is 0.002. Across every plotted parameter, the largest difference is 0.003: Monte Carlo error at the scale of the plot.

What about going from sum-to-zero to population models?

Let s_α denote the prior SD of α_{bayes} and set $v_m = \tau^2 / J$. Because $\alpha_{\text{s2z}} = \alpha_{\text{bayes}} + \bar{a}$,

$$w = \frac{v_m}{s_\alpha^2 + v_m}, \quad (1)$$

$$v_c = \frac{s_\alpha^2 v_m}{s_\alpha^2 + v_m}, \quad (2)$$

$$\bar{a} \mid \alpha_{\text{s2z}}, \tau \sim \mathcal{N}(w \alpha_{\text{s2z}}, v_c). \quad (3)$$

For every sum-to-zero posterior draw:

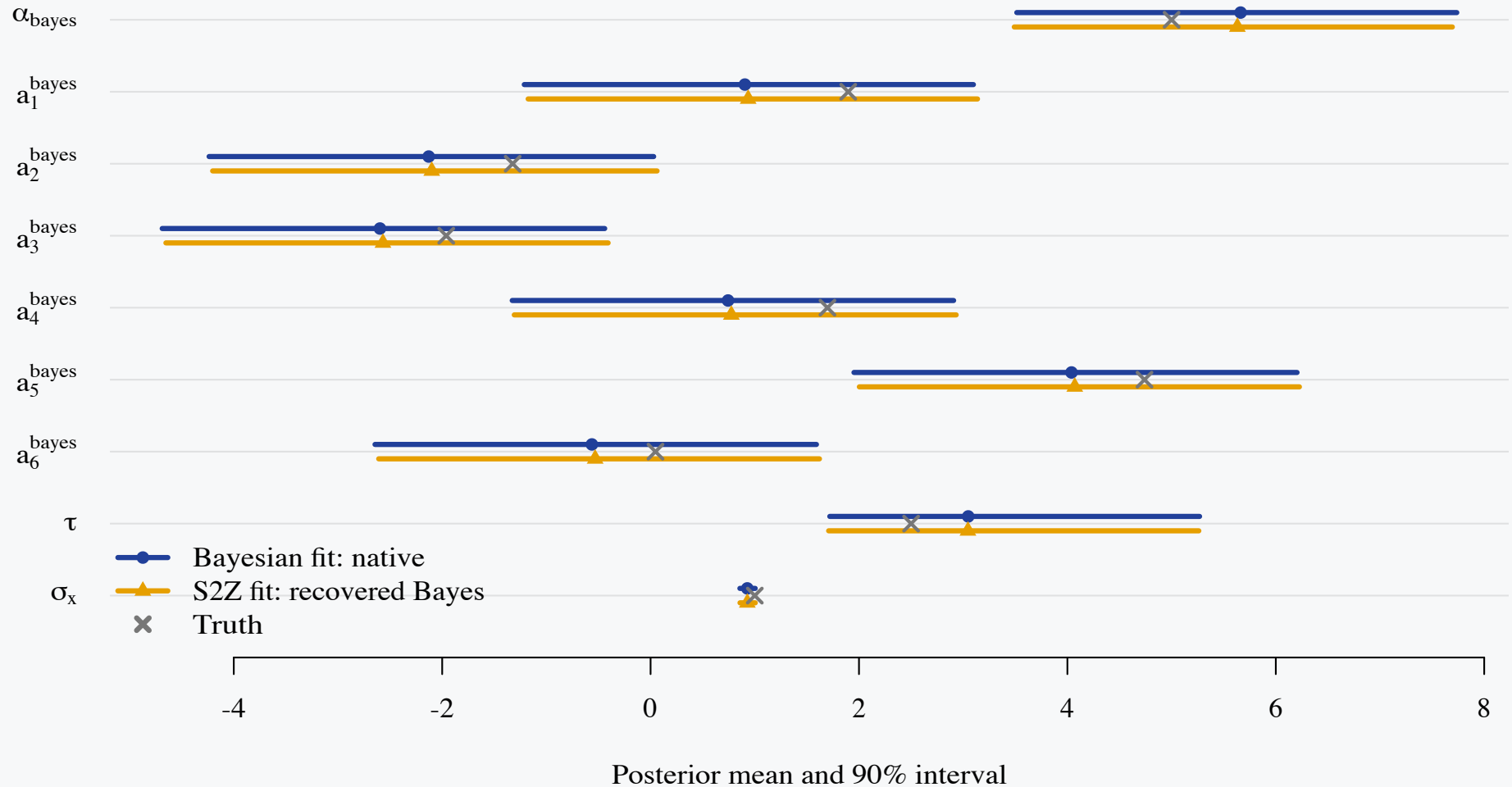
1. Sample $m \sim \mathcal{N}(w \alpha_{\text{s2z}}, v_c)$, using (mean, variance) notation.
2. Set $\alpha_{\text{bayes}} = \alpha_{\text{s2z}} - m$ and $a_g^{\text{bayes}} = a_g^{\text{s2z}} + m$.

Sum-to-zero specification and conversion

The sum-to-zero parameter lives in $J - 1$ subspace a normal prior divides by J instead of $J - 1$, add back a $\log(\tau)$

```
1 parameters {
2   real alpha_s2z; real<lower=0> tau, sigma_x;
3   sum_to_zero_vector[J] a_s2z;
4 }
5 model {
6   tau ~ normal(0, 5); sigma_x ~ normal(0, 2);
7   alpha_s2z ~ normal(0, hypot(alpha_prior_sd, tau / sqrt(J)));
8   a_s2z ~ normal(0, tau);
9   target += log(tau);
10  x ~ normal(alpha_s2z + a_s2z[group], sigma_x);
11 }
12 generated quantities {
13   real mean_a_bayes, alpha_bayes_recovered;
14   vector[J] a_bayes_recovered;
15   {
16     real var_mean_a = square(tau) / J;
17     real var_alpha = square(alpha_prior_sd);
18     real weight = var_mean_a / (var_alpha + var_mean_a);
19     real conditional_sd = sqrt(var_alpha * var_mean_a / (var_alpha + var_mean_a));
20     mean_a_bayes = normal_rng(weight * alpha_s2z, conditional_sd);
21   }
22   alpha_bayes_recovered = alpha_s2z - mean_a_bayes;
```

Back to Bayesian: the two fits agree



Across α_{bayes} and the six a_j^{bayes} , the largest posterior-mean difference is 0.032. Across every plotted parameter, the largest difference is 0.032. The largest standardized gap is 1.6 combined Monte Carlo SEs.

On the difference between two posterior means

For each parameter, we calculate

$$z = \frac{|\hat{\mu}_{\text{Bayes}} - \hat{\mu}_{\text{S2Z}}|}{\sqrt{\text{MCSE}_{\text{Bayes}}^2 + \text{MCSE}_{\text{S2Z}}^2}}.$$

The largest value was 1.6. So the worst observed difference was only 1.6 times the noise expected from running two independent finite MCMC simulations. A standard-normal fluctuation exceeds 1.6 in absolute value about 11% of the time.

ESS and ESS/grad on the two parameterizations

Parameter	Bayesian		Sum-to-zero	
	ESS	ESS/grad	ESS	ESS/grad
α	5,557	0.002	33,461	0.121
a_1	5,719	0.002	39,725	0.144
a_2	5,683	0.002	64,383	0.234
a_3	5,650	0.002	71,686	0.260
a_4	5,588	0.002	56,372	0.204
a_5	5,581	0.002	46,215	0.168
a_6	5,572	0.002	39,120	0.142
τ	11,574	0.004	57,662	0.209
σ_x	16,549	0.006	61,344	0.223

4 chains $\times$ 10,000 sampling iterations = 40,000 post-warmup draws, plus 4,000 warmup iterations per chain. Bulk ESS is pooled across chains. ESS/grad divides by the total post-warmup `n_leapfrog__`: Bayesian = 2,796,128; sum-to-zero = 275,672.

Different variances on the varying intercepts

Let $m = \bar{a}_{\text{bayes}}$, $s = \alpha_{s2z}$, and $z_j = a_j^{s2z}$, with $\sum_j z_j = 0$. We sample (s, z, m) but place the priors on the recovered coordinates

$$\alpha_{\text{bayes}} = s - m, \quad a_j^{\text{bayes}} = z_j + m, \quad a_j^{\text{bayes}} \sim \mathcal{N}(0, \sigma_j^2).$$

Consequently, the recovered priors already contribute the following quadratic in m :

$$\log p(m \mid s, \mathbf{z}) = C - \frac{1}{2} \left\{ \frac{(s - m)^2}{s_\alpha^2} + \sum_{j=1}^J \frac{(z_j + m)^2}{\sigma_j^2} \right\}.$$

Writing $\lambda = s_\alpha^{-2} + \sum_j \sigma_j^{-2}$ gives

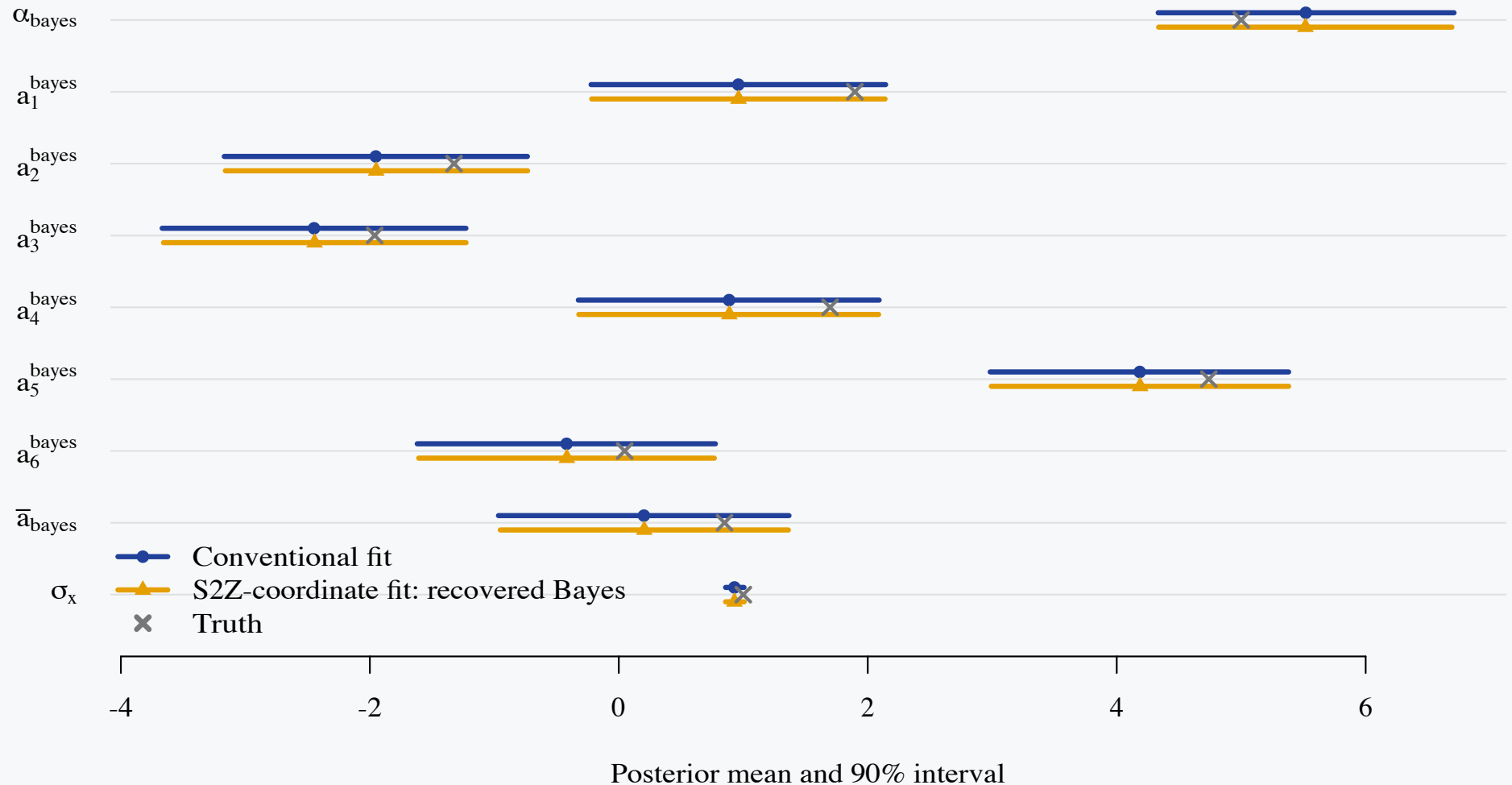
$$m \mid s, \mathbf{z} \sim \mathcal{N} \left(\frac{s/s_\alpha^2 - \sum_j z_j/\sigma_j^2}{\lambda}, \frac{1}{\lambda} \right), \quad m \sim \mathcal{N} \left(0, \frac{1}{J^2} \sum_j \sigma_j^2 \right).$$

It is the correct induced prior on m , so no need to place a prior directly on m .

Stan: different prior SDs

```
1 data {
2   int<lower=1> N;
3   int<lower=2> J;
4   array[N] int<lower=1, upper=J> group;
5   vector[N] x;
6   real<lower=0> alpha_prior_sd;
7   vector<lower=0>[J] a_prior_sd;
8 }
9 parameters {
10  real alpha_s2z;
11  sum_to_zero_vector[J] a_s2z;
12  real mean_a_bayes;
13  real<lower=0> sigma_x;
14 }
15 transformed parameters {
16  real alpha_bayes_recovered = alpha_s2z - mean_a_bayes;
17  vector[J] a_bayes_recovered = a_s2z + mean_a_bayes;
18 }
19 model {
20  sigma_x ~ normal(0, 2);
21  // Priors are placed in the intuitive Bayesian coordinates.
22  alpha_bayes_recovered ~ normal(0, alpha_prior_sd);
23  a_bayes_recovered ~ normal(0, a_prior_sd);
24  // Exact change of variables; constant because J is fixed.
25  target += 0.5 * log(J);
26  x ~ normal(alpha_bayes_recovered + a_bayes_recovered[group], sigma_x);
27 }
```

Different variances: the two fits agree



The six prior SDs are 1, 1.5, 2, 2.5, 3, 3.5. Across every plotted parameter, the largest posterior-mean difference is 0.004 and the largest standardized gap is 0.4 combined Monte Carlo SEs.

Sampling efficiency

Parameter	Conventional		S2Z coordinates	
	ESS	ESS/grad	ESS	ESS/grad
α	6,263	0.004	61,647	0.220
a_1	7,061	0.005	70,771	0.253
a_2	6,783	0.004	68,576	0.245
a_3	6,560	0.004	64,527	0.231
a_4	6,510	0.004	63,248	0.226
a_5	6,405	0.004	61,533	0.220
a_6	6,332	0.004	60,586	0.217
$\bar{a}$	6,225	0.004	66,282	0.237
σ_x	15,363	0.010	61,945	0.221

Matched Bayesian estimands from 40,000 post-warmup draws. ESS/grad divides by total post-warmup `n_leapfrog__`: conventional = 1,549,108; S2Z coordinates = 279,784.

Take-away

Our Bayesian hierarchical models are doing something special.

Treating the varying effects as random variables allows specifying a distribution. For elliptical distributions we can convert across the two parameterizations!

Don't be afraid to use sum-to-zero constraints in your models.

More to come in the paper

- Multiple sets of varying effects
- Varying slopes conversions

Thank you!



github.com/spinkney/open-talks

Delmas, Jean-Pierre. 2024. “Background on Real and Complex Elliptically Symmetric Distributions.” In *Elliptically Symmetric Distributions in Signal Processing and Machine Learning*, edited by Jean-Pierre Delmas, Mohammed Nabil El Korso, Stefano Fortunati, and Frédéric Pascal, 1–34. Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-52116-4_1.